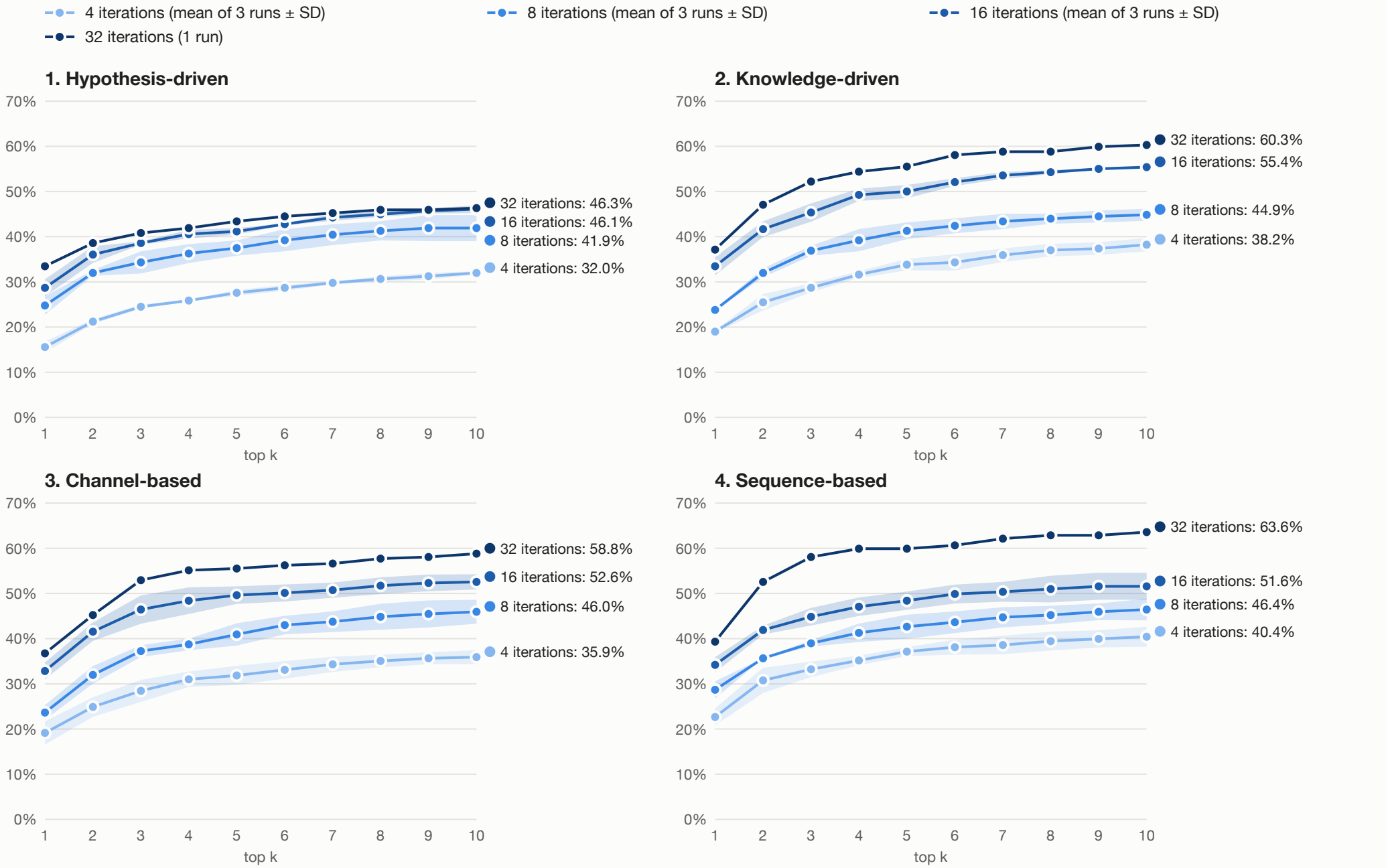
Top-k accuracy by number of iterations

272 cases per strategy and run. Top k: the true diagnosis's ICD-11 code is among the strategy's first k diagnoses (exact code match). Lines are the mean of repeated runs (4 iterations: 3, 8 iterations: 3, 16 iterations: 3, 32 iterations: 1); the shaded band is ±1 sample standard deviation across them.


4 iterations (mean of 3 runs ± SD)
8 iterations (mean of 3 runs ± SD)
16 iterations (mean of 3 runs ± SD)
32 iterations (1 run)
1. Hypothesis-driven
70%
60%
50%
40%
30%
20%
10%
0%
1 2 3 4 5 6 7 8 9 10
top k
32 iterations: 46.3%
16 iterations: 46.1%
8 iterations: 41.9%
4 iterations: 32.0%
2. Knowledge-driven
70%
60%
50%
40%
30%
20%
10%
0%
1 2 3 4 5 6 7 8 9 10
top k
32 iterations: 60.3%
16 iterations: 55.4%
8 iterations: 44.9%
4 iterations: 38.2%
3. Channel-based
70%
60%
50%
40%
30%
20%
10%
0%
1 2 3 4 5 6 7 8 9 10
top k
32 iterations: 58.8%
16 iterations: 52.6%
8 iterations: 46.0%
4 iterations: 35.9%
4. Sequence-based
70%
60%
50%
40%
30%
20%
10%
0%
1 2 3 4 5 6 7 8 9 10
top k
32 iterations: 63.6%
16 iterations: 51.6%
8 iterations: 46.4%
4 iterations: 40.4%

Strategy	Run	top 1	top 2	top 3	top 4	top 5	top 6	top 7	top 8	top 9	top 10
1. Hypothesis-driven	4 iterations	15.6 ± 1.2	21.2 ± 0.6	24.5 ± 0.4	25.9 ± 0.2	27.6 ± 0.6	28.7 ± 0.6	29.8 ± 0.4	30.6 ± 0.6	31.2 ± 0.7	32.0 ± 0.4
1. Hypothesis-driven	8 iterations	24.8 ± 2.0	32.0 ± 0.6	34.3 ± 2.4	36.3 ± 2.1	37.5 ± 1.7	39.2 ± 2.4	40.4 ± 2.3	41.3 ± 2.1	41.9 ± 2.9	41.9 ± 2.9
1. Hypothesis-driven	16 iterations	28.7 ± 1.9	36.0 ± 1.8	38.6 ± 0.4	40.6 ± 0.9	41.2 ± 0.7	42.8 ± 0.2	44.2 ± 0.8	45.0 ± 0.6	45.7 ± 0.6	46.1 ± 0.8
1. Hypothesis-driven	32 iterations	33.5	38.6	40.8	41.9	43.4	44.5	45.2	46.0	46.0	46.3
2. Knowledge-driven	4 iterations	19.0 ± 0.2	25.5 ± 1.8	28.7 ± 1.0	31.6 ± 0.7	33.8 ± 1.3	34.3 ± 1.8	35.9 ± 1.5	37.0 ± 1.4	37.4 ± 1.4	38.2 ± 1.5
2. Knowledge-driven	8 iterations	23.8 ± 0.2	32.0 ± 1.0	36.9 ± 1.1	39.2 ± 2.4	41.3 ± 1.9	42.4 ± 1.7	43.4 ± 1.7	44.0 ± 1.1	44.5 ± 1.3	44.9 ± 1.3
2. Knowledge-driven	16 iterations	33.5 ± 2.0	41.7 ± 1.7	45.3 ± 2.0	49.3 ± 1.3	50.0 ± 1.5	52.1 ± 0.8	53.6 ± 0.8	54.3 ± 0.4	55.0 ± 0.2	55.4 ± 0.2
2. Knowledge-driven	32 iterations	37.1	47.1	52.2	54.4	55.5	58.1	58.8	58.8	59.9	60.3
3. Channel-based	4 iterations	19.1 ± 2.6	24.9 ± 2.2	28.4 ± 2.4	31.0 ± 1.7	31.9 ± 2.1	33.1 ± 1.9	34.3 ± 1.7	35.0 ± 1.4	35.7 ± 1.3	35.9 ± 1.5
3. Channel-based	8 iterations	23.7 ± 1.7	32.0 ± 1.9	37.3 ± 1.3	38.7 ± 1.3	40.9 ± 2.4	43.0 ± 2.0	43.8 ± 2.3	44.9 ± 2.9	45.5 ± 3.0	46.0 ± 2.7
3. Channel-based	16 iterations	32.8 ± 1.9	41.5 ± 2.0	46.4 ± 3.1	48.4 ± 2.9	49.6 ± 1.9	50.1 ± 1.9	50.7 ± 1.7	51.7 ± 1.9	52.3 ± 1.9	52.6 ± 1.7
3. Channel-based	32 iterations	36.8	45.2	52.9	55.1	55.5	56.2	56.6	57.7	58.1	58.8
4. Sequence-based	4 iterations	22.7 ± 1.9	30.8 ± 2.8	33.2 ± 1.7	35.2 ± 1.1	37.1 ± 1.0	38.1 ± 1.7	38.6 ± 2.0	39.5 ± 2.1	40.0 ± 1.9	40.4 ± 2.2
4. Sequence-based	8 iterations	28.7 ± 1.9	35.7 ± 0.0	39.0 ± 0.7	41.3 ± 2.0	42.6 ± 2.7	43.6 ± 2.4	44.7 ± 2.2	45.2 ± 2.0	46.0 ± 1.8	46.4 ± 2.4
4. Sequence-based	16 iterations	34.2 ± 1.7	41.9 ± 1.0	44.9 ± 1.9	47.1 ± 2.0	48.4 ± 2.0	49.9 ± 2.1	50.4 ± 2.2	51.0 ± 2.9	51.6 ± 3.0	51.6 ± 3.0
4. Sequence-based	32 iterations	39.3	52.6	58.1	59.9	59.9	60.7	62.1	62.9	62.9	63.6